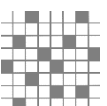


V5095

Controller Board

Document Revision: 1.0

Technical Reference

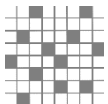


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Revision history

Revision	Description	Date
1.0	Initial Release (Preliminary)	2025-02-10

Introduction

This document describes how to connect and operate the ViALUX V5095 DLP® Controller Board of UltraSpeed V-Modules.

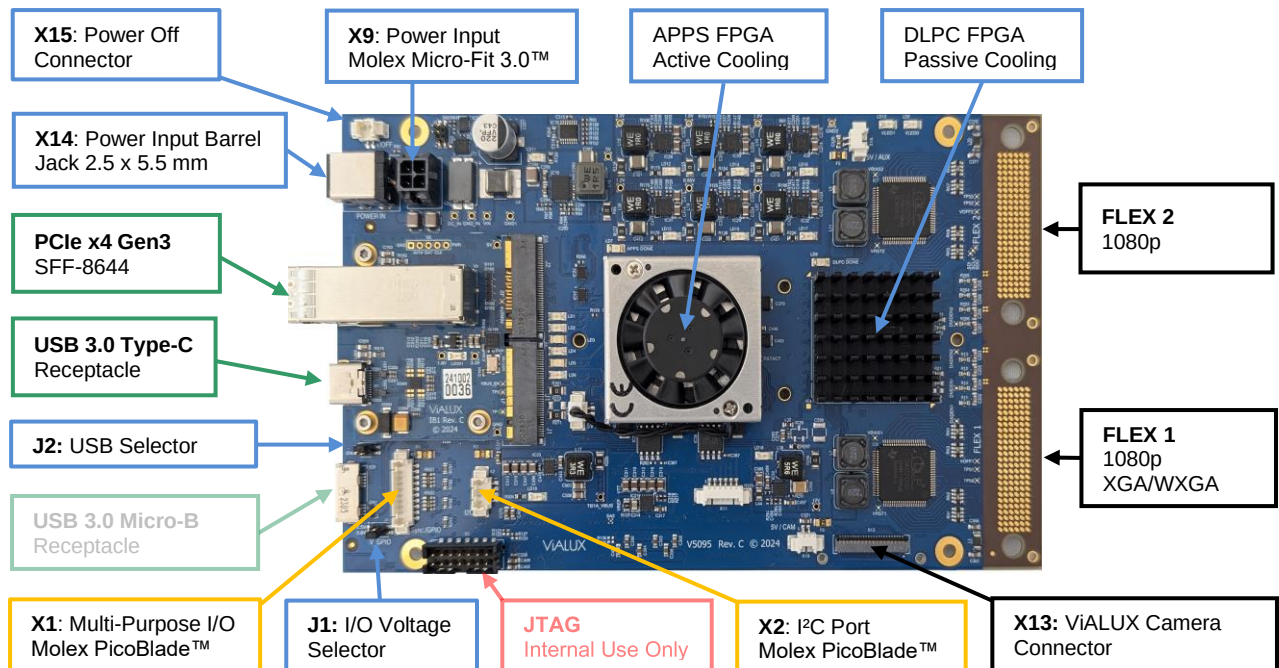
The DLP® V5095 Controller Board provides control logic and power supply for driving one of the following DMDs using Texas Instruments DLPC410 chipset:

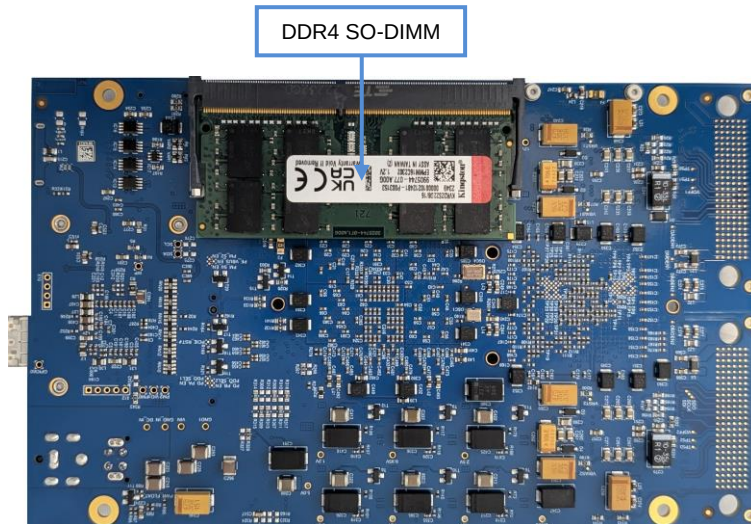
- 0.95" 1080p 1920x1080 (DLP9500/UV)
- 0.70" XGA 1024x768 (DLP7000/UV)
- 0.65" WXGA 1280x800 (DLP650LNIR)

Key features of the V5095 Controller Board are

- PCIe x4 Gen3 PC interface
- USB 3.0 PC interface
- DDR4 SDRAM with up to 32GB on-board memory

Connectors





SO-DIMM Socket

A standard DDR4 SO-DIMM (Small Outline Dual-Inline Memory Module, 260 pins) is used for image storage. Please use the module specified and supplied by ViALUX only. Always switch off the device before removal or insertion of the memory module.

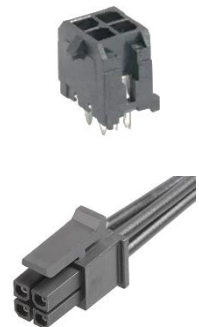
Power Connectors

A 4-pin Molex DC power jack or, alternatively, a Barrel Receptacle is provided on the V5095 to connect to a power supply unit. Please do **not** use both power connectors simultaneously.

X9: Molex Micro-Fit 3.0™

This is a Molex Micro-Fit 3.0™ dual row header, Molex part number 43045-0424. It mates with receptacle 43025-0400 or cable assembly 2147561043.

Pin Number	Signal Name	Power Consumption
3 – 4	24V	2.5A
1 – 2	GND	



X14: DC Power Barrel Jack

Center pin diameter 2.5 mm, outside diameter 5.5 mm, Insertion Depth 9 mm

Pin	Signal Name	Power Consumption
Inside	24V	2.5A
Outside	GND	



Use the barrel connector alternatively only if X9 is not in use. For production environment we recommend usage of X9 due to its interlock mechanism.

PCIe x4 Gen3 Connector (SFF-8644)

This is the connector for PCIe connection to the PC. Use the supplied PCIe cable and the supplied PCIe adapter card to connect to the board. Refer to the [PCIe Adapter Cards User Guide.pdf](#) for detailed installations instructions.



USB 3.0 Type-C Connector

This is the connector for USB connection to the PC. Use the supplied USB cable to connect to the board.



USB 3.0 Micro-B Connector (Legacy)

This is the legacy USB Micro-B connector for backward compatibility with the ViALUX V4395 DLP® Controller Board. By the default this connector is disabled. Use jumper J2 to enable the USB Micro-B connector and disable the USB Type-C connector. See Jumper J2 description below for details.



Jumper J2: USB Selector

Jumper J2 is used to select whether the USB Type-C connector (Default) or the legacy USB Micro-B connector is activated.

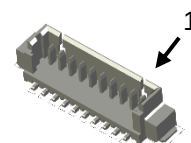
Position	USB-Type-C	USB Micro-B
Open	Enabled	Disabled
1 – 2	Disabled	Enabled
3 – 4 (Park Position)	Enabled	Disabled



X1: Multi-Purpose I/O Connector

The connector is a Molex PicoBlade™ header, part number 53398-1071. It mates with Molex part number 51021-1000. Use crimp contacts 50058-8000 or 50079-8000.

For your convenience the package includes a set of prepared cables and a connector housing.



This connector is not hot-plug capable. Do not attach or detach the plug while the device is powered in order to avoid damage. Always apply a common GND when connecting to other devices.

The table below gives default directions and usage of pins after ALP API initialization.

The ALP-5 API can change signal direction for whole groups only (A – D). For details and other usage please refer to the ALP-5 API description.

On power-up, all signal pins (2 to 9) are configured as inputs.

Pin	Signal Name	Direction	Group	Purpose
1	VCC_GPIO	IN/OUT		I/O Voltage (See configuration of jumper J1)
2	SYNCH_OUT1_GATE	OUT	A	Dynamic Gate Synchronisation Output 1 (Refer to the ALP-5 API description) ALP_DEV_DYN_SYNCH_OUT1_GATE
3	SYNCH_OUT2_GATE	OUT	A	ALP_DEV_DYN_SYNCH_OUT2_GATE
4	SYNCH_OUT3_GATE	OUT	B	ALP_DEV_DYN_SYNCH_OUT3_GATE
5		OUT	B	Reserved
6		IN	C	Reserved
7	TRIGGER_IN	IN	C	Trigger Input
8	SYNCH_OUT	OUT	D	Frame Synchronization Output
9	PWM	OUT	D	ALP_PWM_LEVEL: Output a pulse-width modulated value
10	GND	–		Ground

Weak pull-up resistors (4k7-10k to VCC_GPIO) on all signals provide a defined level for each pin when floating (input). In order to satisfy operating conditions, do not add external pull-down resistors. For details refer to the datasheet of SN74LVC2T45 and the schematic below.

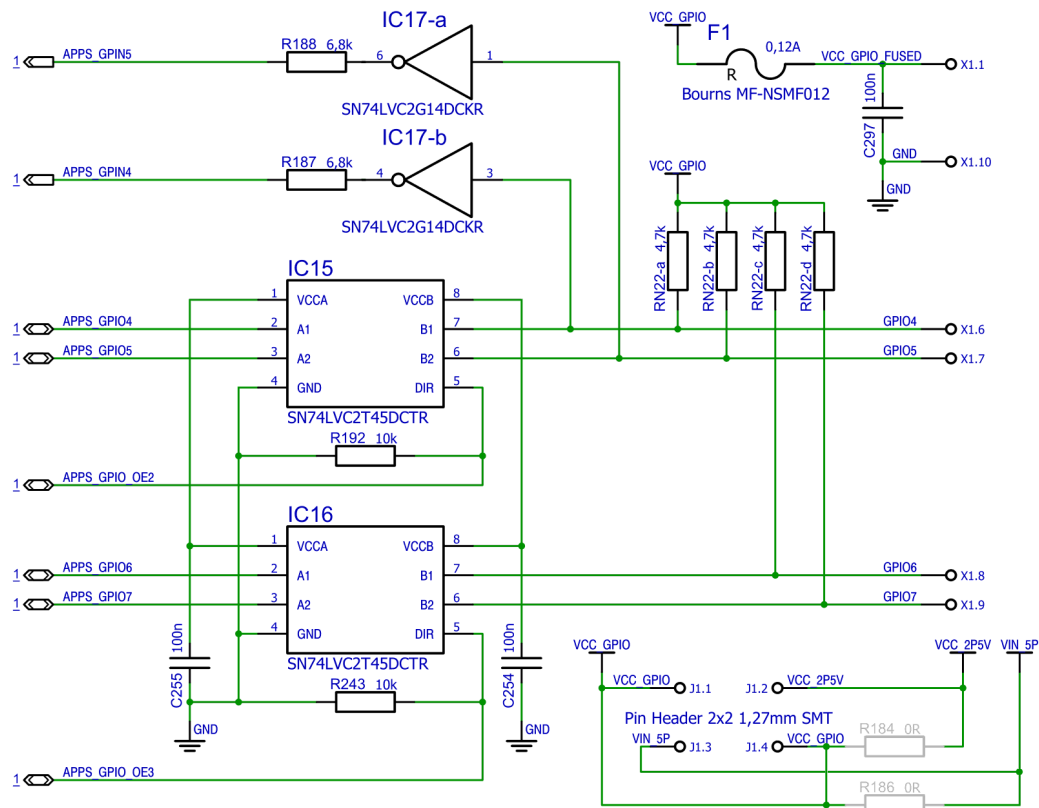
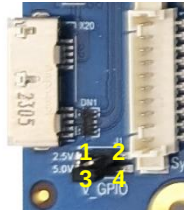
Additional Schmitt Triggers SN74LVC2G14 at pins 6 and 7 improve robustness when configured as input.

Jumper J1: I/O Voltage Selection

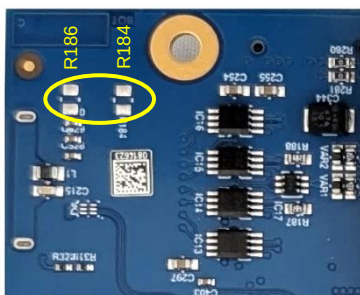
Jumper J1 is used to select either 2.5 V or 5.0 V voltage supply for the multi-purpose I/O connector.

The attached hardware can supply the voltage level, when the jumper is left open. VCC_GPIO is connected to pin 1 of X1 (Multi-Purpose I/O) via a resettable fuse.

Position	0R (0603)	I/O voltage	VCC_GPIO Source
Open	-	2.5 V to 5.0 V	External (X1.1)
1 – 2	R184	2.5 V	Internal
3 – 4	R186	5.0 V	Internal



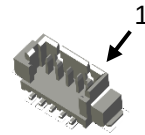
For use in high vibrational environments, it is recommended to substitute the jumper by a single 0R resistor (0603). Positions R184 and R186 are located on bottom side below J1:



Always use one jumper only or alternatively one single resistor in order to avoid board damage by a short circuit between both VCC sources.

X2: I2C connector

This connector is a Molex PicoBlade™ header, part number 53398-0471. It mates with Molex part number 51021-0400, use crimp contacts 50058-8000 or 50079-8000.



This connector is not hot-plug capable. Do not attach or detach a plug while the device is powered in order to avoid damage. Always apply a common GND when connecting devices.

Pin	Signal Name	Direction	Purpose
1	GND		Ground
2	VCC_3P3V	OUT	Power (3.3V, max. 120mA, fused)
3	SDA	IN/OUT	Serial Data
4	SCL	OUT	Serial Clock

LED Description

Table 1 lists the on-board LEDs with its functional description.

Table 1 – Description of LEDs

LED	Color	Status	Description
LD1	Green	Steady	APPS FPGA initialization complete
LD2	Green	Steady	DLPC FPGA initialization complete
LD3	Blue	Reserved	
LD4	Green	Reserved	
LD5	Blue	Steady	PCIe connected
		Blinking	PCIe transferring data
LD6	Green	Steady	PCIe link is 8.0 GT/s (Gen3)
		Blinking (2 Hz, 50% duty cycle)	PCIe link is 5.0 GT/s (Gen2)
		Blinking (1 Hz, 50% duty cycle)	PCIe link is 2.5 GT/s (Gen1)
LD7	Red	Steady	APPS_DONE: FPGA not configured
LD8	Red	Steady	DLPC_DONE: FPGA not configured
LD9	Green	Blinking (1 Hz, 50% duty cycle)	VLED0: DLPC system heartbeat
LD10	Green	Steady	VLED1: DLPC initialization complete
LD11	Green	Steady	5V power regulator output
LD12	Green	Steady	3.3V power regulator output
LD13	Green	Steady	1.2V power regulator output
LD14	Green	Steady	1.8V power regulator output

LD15	Green	Steady	0.95V power regulator output
LD16	Green	Steady	1.0V power regulator output
LD17	Green	Steady	2.5V power regulator output
LD18	Green	Steady	12V power regulator output
LD19	Green	Steady	PD power regulator

Protection Features

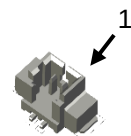
The V5095 board provides protection logic control for the DMD as well as for input voltage and polarity.

Power Float

In the event of power failure or if the power supply is shut down, the DMD will be issued a FLOAT command that places all mirrors in a non-deflected position. The threshold is below 14 V. The ALP-5 API implements ALP_DEV_DMD_MODE for checking this state.

X13: Power Off Connector

This connector is a Molex PicoBlade™ header, part number 53398-0271. It mates with Molex part number 51021-0200. Use crimp contacts 50058-8000 or 50079-8000.



This connector is not hot-plug capable. Do not attach or detach a plug while the device is powered in order to avoid damage.

Pin	Signal Name	Purpose
1	POWER OFF	Connect to GND to shut down the board
2	GND	Ground

If not used, the board is activated as soon as the power supply is available. With the power off signal you can shut down the regulators on the board. In this case you will have a standby current consumption of about 300 μ A.

Operating Conditions

Ambient Conditions		
Storage Temperature	-40 ... 80 °C	
Relative Humidity	0 ... 95 % (non-condensing)	
DMD Operating Conditions	See DLP650LNIR , DLP7000 and DLP9500 datasheets	
Maximum Temperature Reading		
The V5095 contains three temperature measurement points that can be read out by the ALP-5 API. Two sensors are located in the FPGA (APPS FPGA, DDC FPGA), and one is on the PCB, below the SO-DIMM. The device is designed to operate at the following temperatures. Note that exceeding this range could result in unreliable operation or device damage.		
Parameter	Value	ALP API InquireType
PCB	10 ... 70 °C	ALP_PCB_TEMPERATURE
APPS FPGA	≤ 80 °C	ALP_APPS_FPGA_TEMPERATURE
DDC FPGA	≤ 80 °C	ALP_DDC_FPGA_TEMPERATURE

Cooling

On the V5095 main board application FPGA requires active cooling while for DDC FPGA cooling a passive solution fits common requirements. As an option, V5095 main board can be delivered without cooling if you intend to apply your own cooling solutions.

DMD Assembly

The DMD is mounted by pressing it into a contact-spring array, so it can be exchanged without soldering. The mechanical force (screw torque) shall be well controlled as given in the table.

Five DMDs in three sizes of housings are supported by the V5095 controller board. The table below refers to the 4x M3 screws in the original mounting hardware of the corresponding V-Modules.

V-Module Type	Torque Specification for V-Module Mounting Set
V-9501	0.45 ± 0.11 Nm (4.0 ± 1.0 inlb)
V-9501 UV	
V-7001	0.34 ± 0.11 Nm (3.0 ± 1.0 inlb)
V-7001 UV	
V-650L	n.a.

WARNING: Custom DMD mounting designs requires careful consideration of the mechanical forces applied to the DMD. The DMD data sheets specify respective limits in section “Systems Mounting Interface Loads”.

Refer to

<https://www.ti.com/product/DLP650LNIR#design-tools-simulation>

<https://www.ti.com/product/DLP7000#design-tools-simulation>

<https://www.ti.com/product/DLP9500#design-tools-simulation>

for understanding the mounting concepts.

Flex Connections

All DMD extension boards are intended to connect V5095 Main board using flex assemblies. 1080p DMDs require two flex connections (FLEX1 + FLEX2), where XGA and WXGA requires only one of them (FLEX1). Three lengths are available:

Part	ViALUX Order Number
Flex assembly with bolster plates, length 5.2", 150 pos.	K-0104
Flex assembly with bolster plates, length 12.2", 150 pos.	K-0105
Flex assembly with bolster plates, length 23.6", 150 pos.	K-0206

Mounting Instructions

Strictly observe ESD protection.

The V5095 Board is designed to work in any mounting position. Good air circulation is recommended.

Never connect DLP6500 or DLP9000(X) DMDs to a V5095 controller board.

Grounding

All mounting holes of the V4395 Main Board are potential free and therefore not grounded or shielded. GND is connected to the power supply input (X6: Pin 1-2, X9: outside).

ATTENTION: All mounting holes of DMD extension boards are connected to ground.

Usually, the V5095 Main Board is driven by a single 24V/6A power supply with GND connection to the safety contact of the 100V...240V AC wall sockets.

Dimension Drawing

See also [V5095 Dimensions.pdf](#)